ARTICLES

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Gating multiple signals through detailed balance of excitation and inhibition in spiking networks

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Recent theoretical work has provided a basic understanding of signal propagation in networks of spiking neurons, but mechanisms for gating and controlling these signals have not been investigated previously. Here we introduce an idea for the gating of multiple signals in cortical networks that combines principles of signal propagation with aspects of balanced networks. Specifically, we studied networks in which incoming excitatory signals are normally cancelled by locally evoked inhibition, leaving the targeted layer unresponsive. Transmission can be gated 'on' by modulating excitatory and inhibitory gains to upset this detailed balance. We illustrate gating through detailed balance in large networks of integrate-and-fire neurons. We show successful gating of multiple signals and study failure modes that produce effects reminiscent of clinically observed pathologies. Provided that the individual signals are detectable, detailed balance has a large capacity for gating multiple signals.

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Experimental observations[1,2] as well as theoretical arguments[3,4] suggest that excitation and inhibition are globally balanced in cortical circuits. In a globally balanced network, each neuron receives large but approximately equal amounts of excitation and inhibition that, on average, cancel each other. Spontaneous activity is driven by fluctuations in the total synaptic input, leading to asynchronous and irregular patterns of spiking[5–8]. Such networks have been used to study signal propagation and to determine conditions that support various signaling schemes[9–16]. However, neurons in these networks are typically only part of a single signaling pathway, and the transmitted signals cannot be gated or rerouted. Cognitive processing requires signal paths to change dynamically according to the information content of the signal and the processing demands of the receiver[17]. This requires precise control and gating of signal-carrying pathways.

We propose a mechanism for gating based on an extension of the concept of globally balanced networks to local cortical circuits in a form that we call 'detailed balance'. Detailed balance implies that, in addition to an overall or global balance, neurons receive equal amounts of excitation and inhibition on subsets of their synaptic inputs that correspond to specific signaling pathways. Activation of a balanced pathway produces little response in the excitatory neurons of the signal-receiving region, but responses can be gated 'on' by a command signal that disrupts the detailed balance. We analyze properties of the resulting gating mechanism and examine some of its failure modes. We show that the mechanism can gate the propagation of signals from multiple different sources to a single group of neurons, and we determine its capacity for gating large numbers of signals.

RESULTS

We explored the idea of detailed balance in a large network of roughly 20,000 integrate-and-fire neurons with both short- and long-range connectivity (**Fig. 1a,b**; Methods). With appropriately adjusted parameters, this network operates in a globally balanced manner, producing irregular, asynchronous activity in the absence of any time-varying or random external input[5–8]. The distribution of firing rates for the network is approximately exponential with an average firing rate per neuron of 8 Hz (**Fig. 1c**), the distribution of average membrane potentials is approximately gaussian with a mean of −60 mV (**Fig. 1d**), the distribution of interspike intervals (ISIs) is broad with peaks reflecting normal firing and bursting (**Fig. 1e**), and the distribution of coefficients of variation for the ISIs is centered at a value slightly greater than 1 (**Fig. 1f**). Average excitatory and inhibitory membrane currents are of approximately equal magnitude and the net current is near zero, indicating the globally balanced state of the network. This network model is intended to provide a sparse representation of the neurons over a fairly large area, not a full description of a single local circuit such as a cortical column.

Signal gating

To investigate signal gating within the network, we embed a two-layered pathway with 'sender' and 'receiver' subnetworks (**Fig. 1b**). These should be viewed as parts of distinct cortical regions. The connections from the sender region are directed to both excitatory and inhibitory neurons in the receiver region. Such targeting to inhibitory interneurons is consistent with data on the specific targeting of long-range excitatory projections to inhibitory interneurons[18]. To generate a signal, we drive the neurons in the sender subnetwork with a set of external Poisson spike trains at various rates. This causes neurons

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E01 缺口陈述：门控机制此前未被研究
作者报告｜作者在摘要首句直接给出研究缺口：脉冲网络的信号传播已有理论认识，但门控与控制机制此前无人研究。这句话界定了本文的贡献边界——它是机制提出与模拟，不是对既有门控模型的比较检验。

E02 动机：通路必须随信息内容动态改变
作者报告｜作者把门控的必要性追溯到认知加工要求：信号通路要随信号内容与接收者需求动态改变。这是把「需要门控」当作前提而非待检验命题来使用，引用时不应把它当作本文验证过的结论。

E04 核心机制：兴奋被局部抑制抵消
作者报告｜作者用一句话概括机制：进入的兴奋性信号通常被局部引发的抑制抵消，使目标层无响应。「通常」二字说明这是默认状态，可被打破——这是全文所有实验条件的设定基础。

E06 适用范围声明：不是单个皮层柱的完整模型
作者报告｜作者主动限定模型的适用范围：它只是较大面积上神经元的一种稀疏表示，不是对单个局部回路（如一个皮层柱）的完整描述。这句话是引用本文结论时最需要一并带上的边界。

E08 平衡态的验证指标：净电流近零
作者报告｜作者用兴奋与抑制膜电流幅度相等、净电流近零来确认全局平衡态成立。这是网络的初始条件而非结果，重要性在于：后续所有「门控」结论都相对于这一被刻意设定好的平衡而言。

Ｅ０３　ｄｅｔａｉｌｅｄ　ｂａｌａｎｃｅ　的定义
作者报告｜这是全文最关键的概念定义：除了整体平衡之外，神经元还要在「对应特定信号通路的突触输入子集」上Ｅ／Ｉ　相等。这一子集条件把「全局平衡」推进为「针对信号的平衡」，也决定了后面的门控必须按通路逐个配置。

Ｅ０５　　设计推论：被平衡的通路几乎不产生响应
作者报告｜作者明确写出这一设计推论：平衡通路被激活时，接收区兴奋神经元几乎没有响应。这使「门控关」可以用「无响应」来操作化，也解释了为什么后面要用相似度而不是发放率来评价门控。

Ｅ０７　　架构假设：信号源同时投射　Ｅ　与　Ｉ

作者报告｜信号源区同时投射到接收区的兴奋性与抑制性神经元，是「ｄｅｔａｉｌｅｄ　ｂａｌａｎｃｅ」得以成立的结构前提。配合方法部分「每个源神经元只投其中一类」的限制可知，实际采用的是更严格的分派，这是本文与真实皮层投射模式偏离之处。

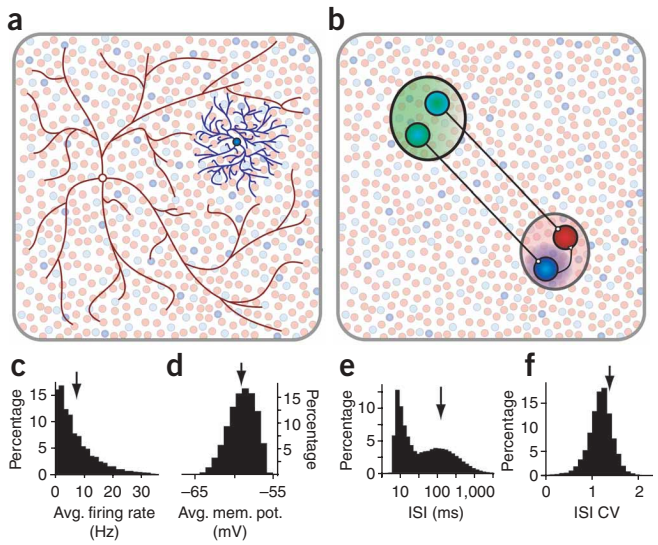


Figure 1 Network connectivity and properties. (**a**) All excitatory and 65% of the inhibitory neurons are connected randomly with a connection probability of 2% (red). The other 35% of the inhibitory neurons (blue) have local connectivity, targeting their nearest neighbors. (**b**) An embedded signal pathway is created by selecting a group of sender neurons (green) that target either excitatory or locally inhibitory neurons (red and blue, respectively, throughout the figures) in a signal-receiving region (red shading) of the network. (**c**–**f**) Asynchronous background activity in the network model. Distributions for network neurons of average (avg.) firing rates (**c**), average membrane potentials (mem. pot.) (**d**), ISIs plotted on a semilog scale (**e**) and coefficients of variation (CV) for those ISIs (**f**). Arrows indicate the means of the distributions.

in the sender area to fire in a manner that mimics the input signal (**Fig. 2a**). In the balanced state, excitation from the sender network, a simple oscillatory signal in the example of **Figure 2**, activates both excitatory and inhibitory subpopulations in the receiver region. The resulting inhibitory activity (blue, **Fig. 2d**) produces a local counter-signal that cancels the excitatory membrane currents (**Fig. 2c**) and generates only modest firing-rate fluctuations in the excitatory receiver neurons (red, **Fig. 2d**,**f**). The signal path is hence gated 'off' in the default (balanced) state of the signal-carrying pathway.

Signal propagation within this network can be gated 'on' in several ways, all of which involve unbalancing the excitatory and inhibitory pathways between the sender and receiver regions. The main requirement is a mechanism that differentially modulates the net excitatory and net inhibitory pathways from the sender to the receiver region[19]. A possible candidate is cholinergic modulation, which satisfies the basic requirements of cell-specific targeting[20,21] as well as relatively rapid response times[22–24]. Rather than modeling such modulation in detail, in the following examples detailed balance is disrupted by decreasing the gain or responsiveness of local inhibitory interneurons in the receiver region. This modulation, in keeping with the strong effects of attention seen for inhibitory neurons[25], corresponds to changing the input–output transfer function so that the same synaptic current generates a smaller response. Although the examples we show focus on modulation of inhibition, any combination of modulations that increases the ratio of excitatory to inhibitory transmission along the signaling pathway will perform similarly. To unbalance the signal in **Figure 2**, we reduced the response gain of the local inhibitory neurons in the receiver region to 15% of its control value. (We discuss more modest gain modulations below.)

Gain modulation that decreases the amplitude of the firing-rate modulations of inhibitory neurons in the receiver region (**Fig. 2d**, blue trace) reduces the inhibition of excitatory neurons in this region, leaving the bulk of the excitatory synaptic current uncanceled (**Fig. 2c**). This produces robust firing in the excitatory receiver neurons that is locked to the temporal pattern of the input signal (**Fig. 2b**,**d**,**f**). Overall network activity is relatively unaffected by these changes (**Fig. 2e**) because the modulated interneurons provide only a small fraction of the total inhibition to the network. Average subthreshold membrane potentials (excluding action potentials and their subsequent refractory periods) of the excitatory receiver neurons in the balanced

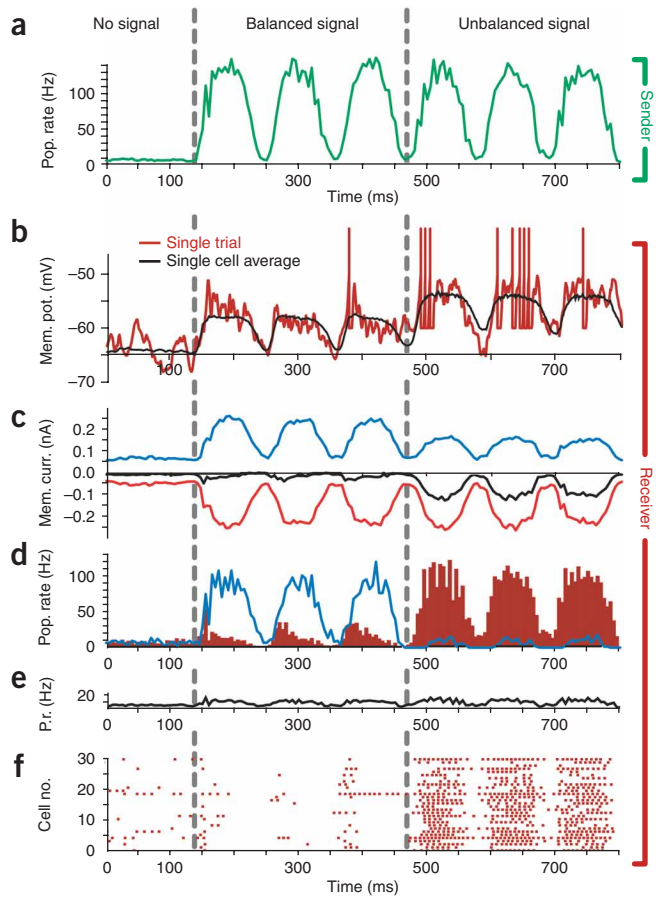


Figure 2 Detailed balance in a network. No signal (left column): all neurons fire at background rates. Balanced signal (middle): sender neurons fire in a correlated manner in response to oscillatory input and project the input firing pattern to both excitatory and inhibitory receiver neurons. Inhibitory receiver neurons reproduce the input pattern, preventing their excitatory neighbors from doing the same. Unbalanced signal (right): by decreasing the responsiveness of the inhibitory receiver neurons, the signal balance in the excitatory receiver neurons shifts in favor of excitation, and the signal is revealed in their firing pattern. All firing rates and averages are calculated in 5-ms bins. (**a**) Population (pop.) average firing rate of the sender neurons responding to a sinusoidally varying input. (**b**) Voltage trace of membrane potential (mem. pot.) in a randomly selected excitatory receiver neuron. Red trace, single trial. Black trace, average subthreshold membrane potential over 100 trials. (**c**) Average membrane currents (curr.) of the excitatory receiver neurons. Red and blue, excitatory and inhibitory currents, respectively; black, net current (including voltage-dependent leak and constant background currents). (**d**) Blue trace, average firing rate of the inhibitory receiver neurons. Red histogram, average firing rate of the excitatory receiver neurons. (**e**) Average population firing rate (p.r.) of the entire network. (**f**) Spike raster for 30 randomly chosen excitatory receiver neurons.

E47　机制的普适性主张：任何提高 E/I 比值的调制均可

AI解读｜作者主张任何提高兴奋与抑制传输比值的调制组合都能起到同样作用，因此本文不依赖某一种特定的生物机制。这是机制可迁移性的最强表述，但它只在本文的 LIF 与手工调参框架内被论证过。

E49　门控不靠压制全网

作者报告｜门控开关几乎不影响整体网络活动，因为被调制的中间神经元只占全网抑制的一小部分。这条结果排除了「门控靠全局压制实现」这一平凡解释，也是该机制具备并行门控能力的前提。

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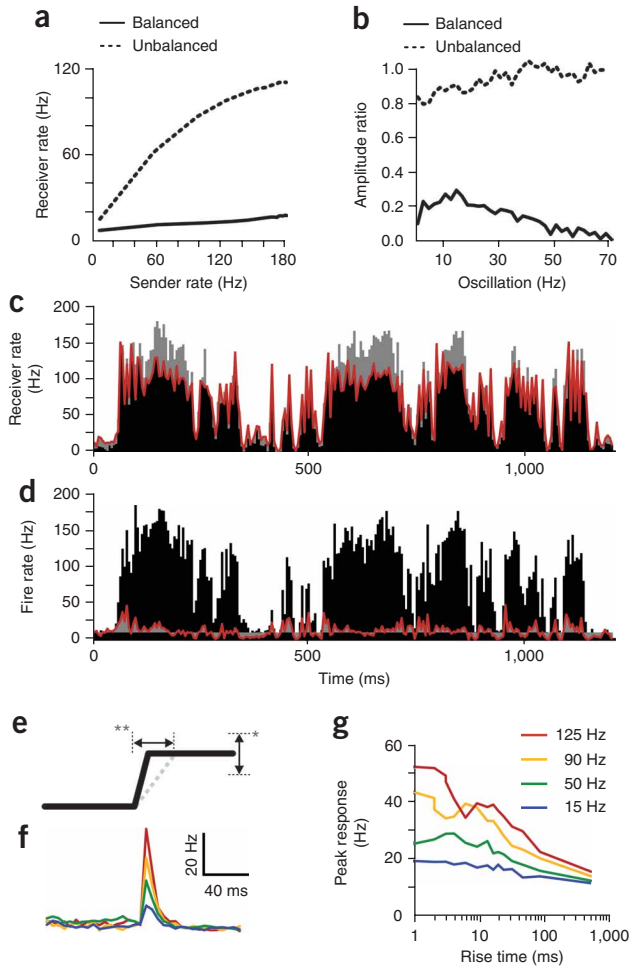


Figure 3 Response analysis. (**a**) Firing rates of the excitatory receiver neurons as a function of different constant sender firing rates, in the balanced (solid trace) and unbalanced (dashed trace) states. (**b**) Ratio of receiver to sender excitatory firing-rate oscillation amplitudes at different oscillation frequencies, in the balanced (solid trace) and unbalanced (dashed trace) states. (**c**,**d**) Response to a random, time-filtered signal in the unbalanced (**c**) and balanced (**d**) states. Red trace, average firing rate of the excitatory receiver neurons; black histogram, rates of the sender neurons. Deviations from the signal in **c** and from the average background rate in **d** are gray. (**e**) Schematic of an input step. Step size (*) and step duration (**) vary independently. (**f**) Average responses of the excitatory receiver neurons in the balanced state to instantaneous steps of different sizes. (**g**) Peak amplitude of the responses in these neurons to steps of different sizes (legend) and rise times (horizontal axis).

and unbalanced states (**Fig. 2b**, black trace) differ by only 4 mV, but this is sufficient to produce markedly different firing patterns.

Response properties

To further quantify the gating mechanism, we studied responses to different types of input (**Fig. 3a**,**b**). The firing rates of excitatory receiver neurons are relatively unaffected by constant input rates in the balanced gated-off state (**Fig. 3a**, solid trace) but rise sharply as a function of input rate when the pathway is gated 'on' (**Fig. 3a**, dashed trace). The rise begins to saturate at high rates because of the residual inhibition produced locally, even at low gain. Gating is also evident in the amplitudes of firing-rate fluctuations for excitatory neurons in the receiver region when the input signal is oscillatory (**Fig. 3b**). In addition, gating occurs when filtered white noise (with a 50-ms time constant[12]) is used as the input signal (**Fig. 3c**,**d**). In the gated-on state, this complex, irregular signal is transmitted with similarity values[12,14] (defined in the Methods) of ~90%, sufficient to propagate the signal across several layers[14]. In the balanced, gated-off state (**Fig. 3d**), the output of the excitatory receiver group is greatly decreased in amplitude, and the similarity between input and output is reduced to ~25%.

A close inspection of the responses in the balanced state (**Fig. 3d**) shows that detailed balance does not completely cancel signals when input firing rates change rapidly. Rapid changes in the signal can evoke a response in the excitatory receiver neurons before inhibition can balance excitation because of the time lag between the monosynaptic excitation and the canceling disynaptic inhibition acting on the receiver neurons. This effect would be even more marked if excitatory and inhibitory synapses were subject to different amounts or types of short-term plasticity. Such partial gating of transients allows large signals with rapid onsets to be propagated, which may induce upstream control circuits to activate gain modulation and open the gate. To further investigate this effect, we activated the sender neurons with step-like input rates of various step sizes and rise times (**Fig. 3e**). Short rise times produce fairly strong responses in excitatory receiver neurons that

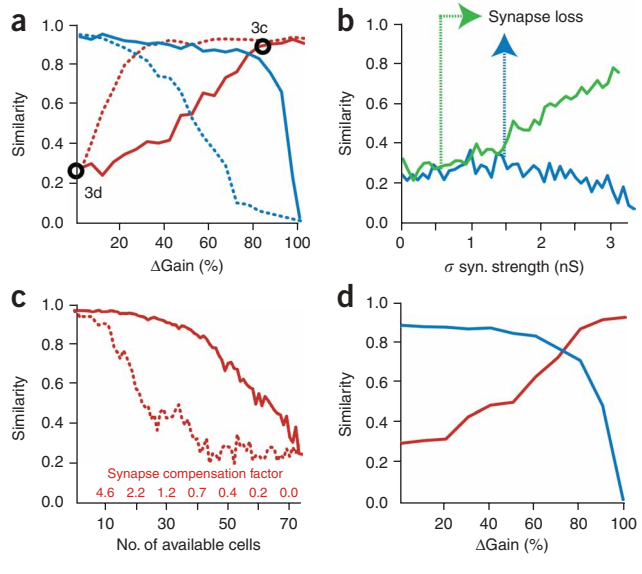


Figure 4 Gain properties. (**a**) Maximum values of the cross-correlations (termed 'similarity'; see Methods) between the sender region and the excitatory (red) and inhibitory (blue) receiver cells for different gains. Solid lines, symmetric gain reduction; dashed lines, asymmetric gain reduction, wherein only the gain of the excitatory synapses onto the inhibitory receiver cells is changed. Circles denote the parameters used in **Figure 3**. (**b**) Similarity values between excitatory receiver activity and the signal in the balanced (gated-off) state as a function of increasing the variability (s.d. σ) of the synaptic (syn.) strengths of the excitatory (green) and inhibitory (blue) pathways. The arrows mark the variability limits beyond which the tails of the strength distributions become rectified to zero. (**c**) Effect of reducing the number of inhibitory receiver neurons on the ability to gate signals off. Similarity values in the balanced state for decreasing numbers of inhibitory receiver cells, without and with synapse strength compensation (solid and dotted lines, respectively). (**d**) Operation of the gating mechanism with only 20 inhibitory receiver neurons by compensating synapse strength and shortened refractory times to allow for more rapid inhibitory firing. Similarity between the signal and the excitatory (red trace) and inhibitory (blue trace) receiver activity is plotted as a function of change in inhibitory gain.

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E19　高输入速率下饱和源于残留抑制
作者报告｜门控打开后接收区发放率随信号源速率急剧上升但在高处饱和，作者归因于局部仍存在的残留抑制。这条限定说明该机制不是无限制的放大器，输入—输出关系仍受抑制残余量支配。

E17　门控打开时的信号保真度约　90%
作者报告｜对滤波白噪声这类不规则信号，门控打开状态下接收区与输入的相似度约　90%。这是本文评估信号可被跨层传递所用的定量标准，与后文容量分析中的相似度指标是同一套。

E50　跨层传播的判据
AI解读｜作者以约　90%　的相似度作为足以让信号跨越若干皮层层级传播的判据，并引既往文献支持。该判据本身沿用他人标准，本文未独立检验跨层传播。

E18　门控关闭时的相似度降到约　25%
作者报告｜门控关闭时输出幅度大幅下降、输入输出相似度降到约　25%。与打开时的约　90%　构成本文门控有效性的核心数值对照，也说明「关闭」并不等于零，仍有约四分之一的残余相关。

E20　关闭状态下的瞬态泄漏
作者报告｜作者细致检查平衡态后指出：输入发放率快速变化时详细平衡并不能完全抵消信号。这是「门控关」并非完全沉默的直接证据，也提示用时间平均发放率评价门控会漏掉这条通路。

E21　泄漏机制：单突触兴奋与双突触抑制的时间差
作者报告｜泄漏的机制被明确指出：单突触兴奋先到、起抵消作用的双突触抑制后到。幅度随阶跃上升时间增加而衰减。作者还指出若兴奋与抑制突触具有不同的短时程可塑性，泄漏会更明显（本文未做该模拟）。

E16　关键数值：膜电位只差　4　mV
作者报告｜这是全文最重要的数值事实：平衡态与失衡态的亚阈值膜电位平均只差　4　mV，发放模式却明显不同。它说明门控是「把已存在的强平衡推过阈值临界点」，而非「改变平均工作点」，因而在实验上应与整体平移类机制区分开。

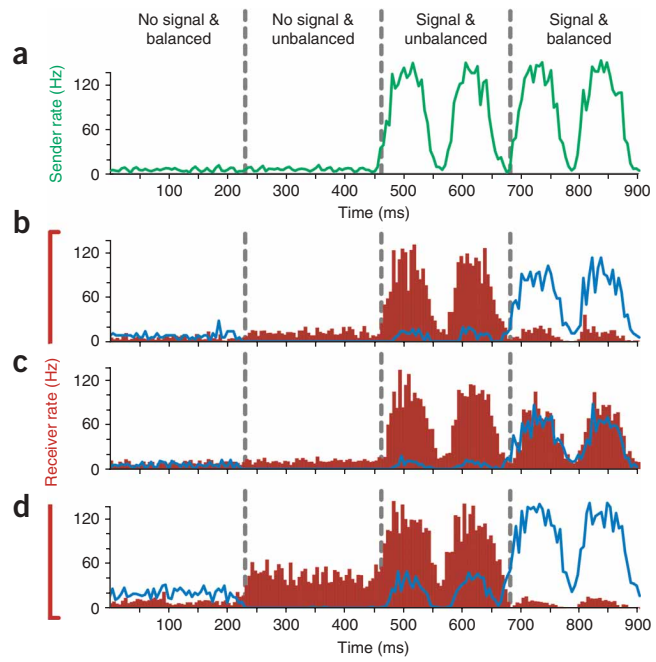


Figure 5 Network pathologies. (**a**) Average firing rate of the sender neurons without and with an oscillatory input. (**b**–**d**) Responses of excitatory (red histogram) and inhibitory (blue trace) receiver neurons with correct tuning (**b**); weakened local inhibition, leading to a gating deficit (**c**); or a hyperactive receiver region causing a response to the gating modulation (**d**). Conditions shown in the different columns are no signal and no modulation, no signal but gated on, signal on and gated on, signal on but gated off. Firing rates are calculated in 5-ms bins.

depend on the step size (**Fig. 3f,g**), even in the balanced gated-off state, but these diminish as the rise time of the step increases, illustrating the transient nature of this transmission.

The gain changes used to gate signals on have been fairly large, so we next examined different degrees and types of gain modulation in the inhibitory receiver neurons. Beginning with no gain change (ΔGain = 0), we decreased the responsiveness and thus the firing rate of the inhibitory receiver population. This causes the firing rate of the excitatory receiver neurons and its similarity to the sender signal in the gated-on state (**Fig. 4a**, solid red trace) to increase. At ΔGain ~80%, the signal similarity of the activity of the inhibitory neurons goes rapidly to zero (**Fig. 4a**, solid blue trace), and the similarity of the excitatory receiver activity plateaus at ~90%. Alternatively, it is possible to reach this same plateau level with a gain shift of only 30% (**Fig. 4a**, dotted traces) by modulating the inhibitory population asymmetrically, which means modifying only the responsiveness to excitatory inputs.

The gating mechanism is robust to many (but not all; see below) different perturbations of the network. To study the effect of synaptic variability, we computed the similarity of responses in the receiver region to the signal, when it is gated off, as a function of the variability in the strengths of the inhibitory (**Fig. 4b**, blue trace) or excitatory (**Fig. 4b**, green trace) sender synapses onto the excitatory receiver cells. Synaptic strength variance does not have a large effect in either case until the variance becomes high enough to force substantial numbers of synapses to zero strength (which occurs at a different point for excitatory and inhibitory synapses because of their different initial strengths), changing the mean synaptic strength. After this point, the high degree of variability in the excitatory synaptic strengths makes it difficult to shut the signal off (**Fig. 4b**).

We also tested the robustness of gating a signal off to the loss of its most critical components, the inhibitory neurons in the receiver region (**Fig. 4c**). The effect of decreasing the number of available interneurons (originally 73) is roughly linear (solid trace), until gating off fails completely when less than 40 cells are available. It is possible to partially rescue gating by upregulating the strengths of all remaining inhibitory synapses proportionately to the number of deleted neurons and thus deleted synapses (dotted trace). However, gating still fails when less than 25 inhibitory cells are available because such a small population of inhibitory neurons cannot fire a sufficient number of action potentials to provide balancing membrane currents, even if their synapses are strengthened to unrealistically high values. If the inhibitory receiver neurons are allowed to spike at rates as high as 600 Hz, it is possible to further rescue the balance mechanism and to successfully gate signals in as many as 600 excitatory cells with as few as 20 inhibitory neurons (**Fig. 4d**).

Pathologies

The basic requirement to achieve a state of detailed balance is local inhibition strong enough to cancel signals in the gated-off state. In addition, the gain modulation used to unbalance and gate 'on' a pathway must not have an excessively destabilizing effect on the global excitatory–inhibitory balance of the network. With this in mind, we examined more ways in which network gating can fail when tuning is relaxed by studying gated off and gated on states with no signal and in the presence of an oscillatory signal (**Fig. 5**). With proper tuning (**Fig. 5b**), the excitatory neurons of the receiver subnetwork respond robustly only when the signal is present and gating is on, although there is a weak transient response when the signal is present but gating is off.

We considered two different detuning conditions. First, we reduced the strength of all synapses from the local inhibitory neurons by 60% (**Fig. 5c**). This causes baseline firing rates in the absence of a signal to rise slightly, but the effect is not large because the bulk of the inhibition is not affected. Little change is seen in the response to the gain modulation alone, but activating the input shows that the gating mechanism has been compromised. Because of the weakened inhibition, excitatory inputs to the excitatory receiver neurons cannot be fully cancelled by local inhibition, and the signal can never be fully gated off.

We also detuned the detailed balance by increasing the strength of excitatory synapses within the receiver area by 60% (**Fig. 5d**). Excitatory synapses onto excitatory and inhibitory neurons were both modulated in this way, so a rough balance is still maintained within the receiver region. As in the case of reduced inhibition, enhanced excitation slightly elevates firing rates in the gated-off, no-signal condition. Activating gain modulation to open the gate causes a substantial elevation in the firing rate of excitatory receiver neurons, even when no signal is present. Thus, with altered excitation, the receiver neurons respond to the gating signal as if it were an input. This means that the network falsely transmits internally generated activity (the gating signal) as if it were an external signal. By contrast, in this condition signal responses in both the gated-on and gated-off states appear normal. We address the implications of these findings in the Discussion.

Multiple signals

One of the advantages of the gating mechanism we propose is that a group of receiver neurons can remain responsive to one set of incoming signals even while other sets are being cancelled by balancing inhibition. This gives the mechanism the capacity to gate multiple signals. As a first example, the network we have been considering is expanded so that it

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E23 失效边界：抑制细胞少于 40 个则关门失败
作者报告｜把接收区抑制性细胞从原 73 个逐步减少，门控效果近似线性下降，但少于 40 个时关门彻底失败。这给出了机制所需的抑制资源量级下限，也是评估其他体系中该机制是否可能存在的第一道筛查。

E24 强化权重也无法挽救的更严格下限
作者报告｜按比例上调其余抑制突触强度可部分挽救，但少于 25 个抑制细胞时仍失败，原因是如此少量的细胞无法发出足够多的动作电位来提供足以抵消的膜电流。这说明该机制对抑制侧的总输出能力有硬性要求。

E25 挽救条件：600 Hz 发放率下 20 个抑制细胞
AI解读｜若允许抑制性接收细胞发放到高达 600 Hz，就能用 20 个抑制细胞门控多达 600 个兴奋细胞。这是一个非生理的发放率，必须与上一条并列引用，否则会高估该机制在生物回路中的可用性。

E22 非对称调制只需 30% 增益位移
作者报告｜若只降低抑制神经元对兴奋输入的响应幅度，30% 的增益位移即可达到对称下调 85% 时的兴奋侧相似度平台。这说明「打开门控」所需的调制幅度可以远小于最直接的实现，为机制提供了较强的可行性余地。

E38 失效模式的判据：门控信号被当成输入
作者报告｜作者对高兴奋失谐的刻画：接收神经元像对待真实输入那样回应门控调制。这是把「控制通道」与「内容通道」混淆的具体表现，也是本文自带的可区分检验——在无信号条件下测接收区即可。

E37 失效模式二：内部活动被误传为外部信号
作者报告｜接收区兴奋被增强 60% 的失谐条件下，网络把内部生成的门控指令当作外部信号传出。这是模型给出的第二种、可与第一种（关不掉）区分开的失效形态，其区分性来自「无信号条件下接收区是否有反应」。

E26 突触强度高变异的鲁棒性上限
作者报告｜突触强度方差在中等范围内不影响门控，但高到迫使大量突触被截断为零、改变平均强度后，兴奋侧的高变异会使信号难以被彻底关闭。这是机制对发育期连接扰动的敏感性提示。

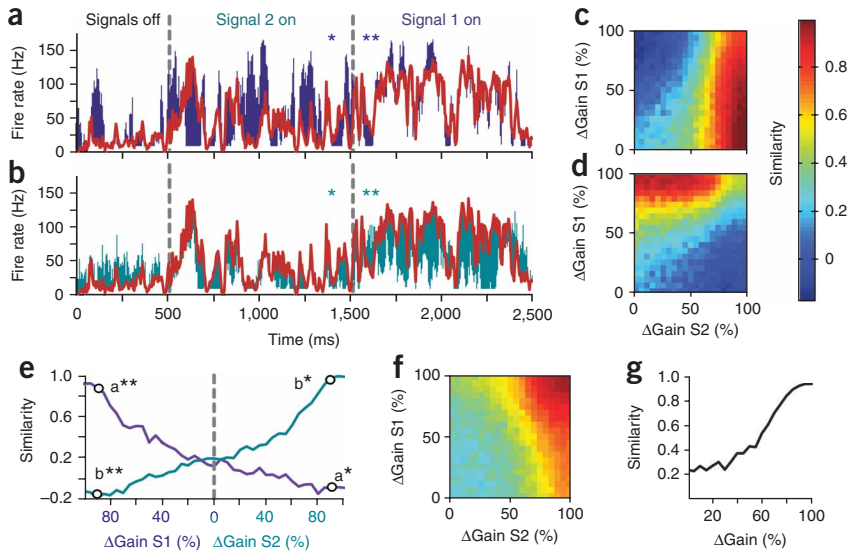


Figure 6 Gating two signals in a network. (**a**,**b**) Average response of the excitatory receiver subnetwork to two simultaneously delivered signals (S1 and S2). Colored areas indicate difference between the average firing rate in the receiver region (red) and either S1 (purple, **a**) or S2 (green, **b**). First column: both signal pathways are balanced, the signals are off. Second and third columns: signal pathways 2 and 1, respectively, are unbalanced by shifting the gain of their respective inhibitory receiver populations to 15% of their control values. (**c**,**d**) Similarity values between S1 and the excitatory receiver activity (**c**) and S2 and the same excitatory receiver activity (**d**) for all possible combinations of the two gain modulations. Both signals reach similarity values of >85%. (**e**) Similarity values for S1 and S2 for independent gain changes. Left of the gray line only the gain for S1 is manipulated while the gain for S2 remains 100%, and vice versa on the right side. Black circles indicate the gain values used for **a**,**b** for the regions denoted by the asterisks. (**f**) Similarity values as in **c**,**d** but measured for the combined signal S1 + S2. (**g**) Similarity values between S1 + S2 and the excitatory receiver subnetwork activity as a function of combined (equal) inhibitory gains, taken from the results along the diagonal of **f**.

can gate two signals, rather than one. We then discuss the capacity of detailed balance for switching a large number of signals using a model that simulates a single neuron in the networks we have been using.

We introduced a second signal to the receiver group. (To accommodate two signals, S1 and S2, we modified the network architecture slightly to allow two separate groups of ~70 inhibitory neurons; see Methods.) We then compared the average firing rate of the excitatory receiver subnetwork compared to each of these signals, S1 and S2 (**Fig. 6a**,**b**). The colored bars indicate the difference between the average firing rate in the receiver region (red) and each of these signals. When both pathways are balanced ('signals off'), firing in the receiver region stays roughly at the background level, except for transient responses as described above. When one of the pathways is unbalanced to allow propagation of its signal, the response of the receiver group follows that signal accurately, as indicated by the small divergence between the appropriate signal and response pair. Activity of the receiver subnetwork does not follow the signal that is in the off state, as indicated by the larger difference regions for the gated-off signals. This finding can also be confirmed using the similarity to S1 and S2, respectively, for all combinations of the two levels of inhibitory gain reduction (**Fig. 6c**,**d**). The regions where similarity is high for either signal are well separated from each other, and both signals reach similarity values above 85% in the regions where they are gated 'on'. Furthermore, when only one of the two signals is gated on, the other signal tends to weaken, and similarity between receiver activity and the gated-off signal can even become negative because of inhibitory overshoot (**Fig. 6e**). To compare the gating of two signals with the gating of one, it is useful to examine the similarity value between the combined signal S1 + S2 and the firing rate of the receiver cells (**Fig. 6f**,**g**). This is much the same as in the single-signal case (**Fig. 4a**, solid red trace).

How many signals can be canceled and then gated 'on' by a population of inhibitory neurons? To address this question, we studied a single excitatory receiver neuron, rather than the full network that we have been considering up to this point. As discussed above for the two-signal case, configuring a network for multiple signals takes a fair amount of modification and readjustment, and this made it unpractical to consider a wide range of different numbers of signals within a full network. Instead, we set up the mechanism of detailed balance in a single integrate-and-fire neuron that receives 800 excitatory and 200 inhibitory inputs. The critical component in determining the capacity of detailed balance for switching multiple signals is the population of inhibitory afferents, because they are less numerous than their excitatory counterparts and must cancel the excitatory effects of the signals while at the same time allowing a particular signal to get through when modulated. We chose 200 inhibitory inputs to match the number that seem to influence a single pyramidal neuron in cortical circuits[26].

The single-neuron model acts much like any of the excitatory receiver neurons in the full network because we adjusted its input to match what a typical neuron receives when the network is intact. The excitatory and inhibitory neurons that provide input to this model neuron are represented by Poisson spike trains generated from firing rates that encode various numbers of signals directly rather than through other model neurons. For this reason, we refer to them as inputs or afferents rather than as neurons—although, of course, they correspond to neurons in the full network. Detailed balance is achieved by distributing the signals across the excitatory and inhibitory afferents and adjusting synaptic strengths so that all signals are cancelled in the default state. To gate a particular signal on, we set the gains of all the inhibitory afferents that carry that signal to zero, essentially shutting them off. We used this extreme form of modulating because we wished to compute the maximum capacity of the system, not a capacity limited by restricting the amount of modulation.

Each signal consisted of a mean firing rate plus independent filtered white noise fluctuations (as used in **Fig. 3c**,**d**). To begin (**Fig. 7**), we distributed M signals across the afferents so that each excitatory and inhibitory afferent carries only one signal. The maximum number of signals that can be distributed in this way is $M = 200$, the number of inhibitory afferents. Performance, measured in terms of a similarity index (see Methods), falls rapidly as a function of the number of signals being gated, and this way of distributing and gating multiple signals does not allow switching of more than ~10 signals (**Fig. 7a**,**d**). The problem is not in canceling out the signals that are not being gated 'on' but in being able to fully gate the chosen signal on.

One problem with the scheme of having one signal per inhibitory afferent is that the number of afferents being gain-modulated to zero to gate a given signal on is small. For example, only 10 inhibitory afferents

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E40 多信号结论的构型依赖
作者报告｜作者承认多信号构型需要大量改造与重新调参才得以建立。这意味着多信号结果是在专门设计并重调过的网络中取得的，其稳健性未经原始调参流程检验，是评估该结论外部效度时的关键限制。

E27 双信号并行门控且选择区彼此分离
作者报告｜在两条信号各自的增益调制二维参数空间中，高相似度区域彼此不相连，两条信号各自在被打开区域都超过85%。这是「通路选择性」而非仅仅「活动增强」的直接证据，也是与背景阅读清单中「门控必须验证通路选择性」这一要求对应的关键结果。

E29 瓶颈定位：不是关不掉，而是打不开
作者报告｜作者对单突触承载单信号方案失效原因的定位很精确：其余信号照样能被关掉，问题在于被选中的信号无法被充分打开（因为承载它的抑制输入太少）。这个诊断直接导向下一步的信号重叠设计。

E28 关闭信号出现负相关（抑制过冲）
作者报告｜只打开一条信号时，另一条被关闭的信号不仅不被跟随，其相似度甚至因抑制过冲变为负值。这说明「关闭」在动力学上带有反向影响，用「不传」这一被动模型无法解释，是区分本文机制与简单抑制模型的一个硬指标。

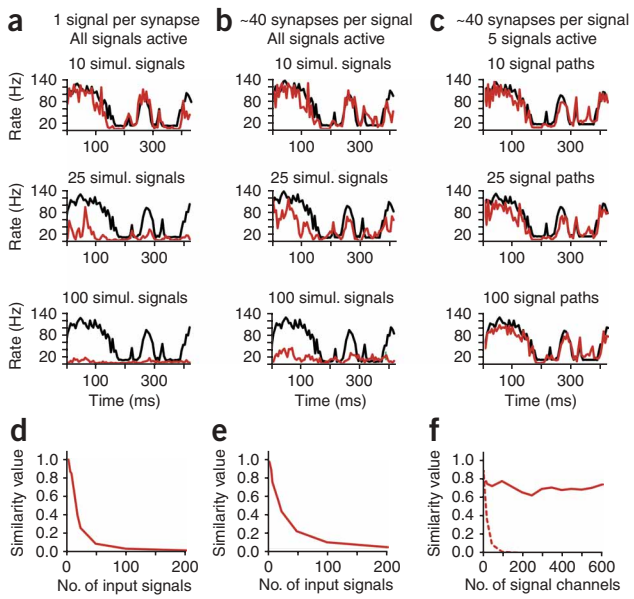


Figure 7 Multiple signals into a single cell. (**a**–**c**) Firing rates of the output signal (red), averaged over 200 runs, compared to the input signal (black). First row: 10 simultaneous (simul.) signal paths. Second row: 25 simultaneous signals paths. Third row: 100 simultaneous signal paths. (**a**) Each afferent to the model neuron carries only one signal. (**b**) Each afferent carries 40 signals. (**c**) Each afferent can carry 40 signals but only 5 signals are present at a given time. (**d**–**f**) Similarity values as a function of the number of signals being gated in **a**–**c**. (**d**) With one signal per afferent, gating is limited to less than about 20 signals. (**e**) Overlapping several signals onto each afferent improves performance slightly. (**f**) When only 5 signals are present at a given time, large numbers of signals can be gated when 40 signals are carried on each afferent (solid trace), but performance is still limited if each afferent carries a single signal (dashed trace).

carry any particular signal when $M = 20$. This problem can be addressed by distributing the M signals across the excitatory and inhibitory afferents so that each afferent carries more than one signal (**Fig. 7b**). Gating works best if each signal is carried on ~40 inhibitory and ~160 excitatory afferents (chosen randomly for each signal from all available afferents of each type), which means that each inhibitory afferent carries, on average, $40M/200$ signals, rather than 1 as before. In this case, because of the overlap in the signals, when a particular signal is gated 'on' by setting the gains of the inhibitory afferents carrying that signal to zero, this upsets the detailed balance for the other, ungated signals. To compensate for this, the gains of the remaining inhibitory afferents are adjusted so that the ungated signals are cancelled as nearly as possible by the remaining active inhibitory afferents. In other words, through a procedure discussed in the Methods, the gains on the remaining active afferents are increased to compensate for those missing owing to gating 'on' of the chosen signal. Performance with this distribution of signals is better than in the one-signal-per-afferent case, but detailed balance still cannot handle more than 30 signals (**Figs. 7b,e**).

What is limiting the ability of the detailed balance scheme to switch large numbers of signals? The limitation is, in fact, not a deficit of the detailed balance scheme but a fundamental problem with encoding multiple signals using firing rates, which no switching scheme can avoid. This is the problem of keeping firing rates positive. As discussed above, each signal corresponds to a mean firing rate plus positive and negative fluctuations about this mean. Although the mean rate carries no information about the signal, it cannot be set to zero or half of the signal would be lost owing to firing-rate rectification. Adding together M signals results in a total input that has a mean proportional to M and a fluctuating signal that is proportional to only $M^{1/2}$ because the M signals are independent of one another. Thus, there is a strong tendency for the mean input to drown out the signals to which we want the postsynaptic neuron to respond[27]. Of course, because of the balancing inhibition, the mean excitatory input is canceled, but this cancellation is subject to fluctuations due to the spiking nature of the inputs. As a result, there is a fundamental limitation in the number of signals from which a single signal can be extracted, independent of the method by which this is done.

To show that the limited capacity (**Fig. 7a,b**) is a result of this fundamental restriction on firing-rate coding and not a limitation of the detailed-balance approach, we restricted the number of signals present on the afferents to the neuron at any given time. In other words, we set up the postsynaptic neuron so that it could extract any one out of M input signals, but at any given time we restricted the number of signals present to 5 out of these M possibilities (**Fig. 7c**). This seems reasonable for an *in vivo* switching situation: out of the myriad of possible stimuli that can activate a neuron, only a few are likely to be present at any given time. The number 5 is arbitrary; the key is to restrict the number of signals at any given time to a value that does not make the signals undetectable owing to the $M^{1/2}$ scaling problem discussed above. When the input signals are restricted in this way, the capacity is still limited when each afferent is only allowed to carry one signal, but when each signal is distributed across 40 afferents, the switching capacity is much larger (**Fig. 7c,f**). There is no decrease in similarity for the gated on signal over the entire range from 1 to 600 possible signals. This shows that the limited performance for the one-signal-per-afferent case (**Fig. 7a**) is a result of having too few afferents per signal. By contrast, the poor performance when 40 afferents are used per signal (**Fig. 7b**) does not represent a limitation of detailed-balance gating but rather a basic limitation of rate-based coding. When this latter limitation is avoided, detailed balance can switch very large numbers of signals.

Table 1 Synapse modifications

		Synapse groups	Healthy	Inhibition deficit	Hyperexcitable receiver	Two signals
Δg_{ex}^{global}	0.8 nS	$\Delta g_{ex\rightarrow ex}^{sender\rightarrow receiver}$	1.1	1.1	1.1	0.9
		$\Delta g_{ex\rightarrow ex}^{net\rightarrow receiver}$	1.0	1.0	1.6	1.0
		$\Delta g_{ex\rightarrow loc\ inh}^{sender\rightarrow receiver\ 1}$	1.0	1.0	1.0	1.1
		$\Delta g_{ex\rightarrow loc\ inh}^{net\rightarrow receiver\ 1}$	1.0	1.0	1.6	1.3
Δg_{inh}^{local}	1.5 nS	All	1.0	0.4	1.0	1.0
		$\Delta g_{loc\ inh\rightarrow ex}^{receiver\ 1\rightarrow receiver}$	3.1	3.1*	3.1	3.3
		$\Delta g_{loc\ inh\rightarrow ex}^{receiver\ 2\rightarrow receiver}$				4.0
Δg_{inh}^{global}	7.5 nS	$\Delta g_{inh\rightarrow loc\ inh}^{net\rightarrow receiver\ 1}$	1.25	1.25	1.25	1.32
		$\Delta g_{inh\rightarrow loc\ inh}^{net\rightarrow receiver\ 2}$				1.2
		$\Delta g_{inh\rightarrow ex}^{net\rightarrow receiver}$				0.75

Overview of synaptic alterations, sorted by synapse type (rows) and purpose of modification (columns). The synaptic strengths appearing in column 2 were multiplied by the factors appearing in columns 4, 5 and 6. The asterisk indicates that this value is 3.1 times the scaled value (scaled by 0.4).

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npg

E39　作者自陈：同时存在的信号数　5　是任意取值
作者报告｜作者明确写出这一取值是任意的，关键只在于把同时存在的信号数限制在不会因　√M　缩放而不可检测的水平。这句话是「可门控数百条信号」这一主张的最重要限定，引用数字时必须一并给出。

E33　绕开编码瓶颈后容量不再受限
作者报告｜把同时存在的信号限制为　5　条后，每突触承载约　40　条信号的方案在　1　至　600　条候选信号范围内相似度都不下降。这一结果支持第　4　步对「600　条」这一数字适用范围所作的范围收窄。

E30　信号重叠后仍不超过　30　条
作者报告｜把每条信号铺在约　40　条抑制与约　160　条兴奋输入上后性能优于单突触方案，但切换能力仍不超过　30　条。这条数字是作者在突破编码瓶颈之前能给出的容量上限。

E31　归因分离：瓶颈在率编码而非门控
作者报告｜作者把容量上限明确归因于用发放率编码多信号这一固有困难，而不是详细平衡方案本身的缺陷。这是本文最重要的方法论贡献之一，把机制瓶颈与编码瓶颈分离开来。

E32　机制解释：均值按　M　增长而涨落按　√M
作者报告｜作者给出容量上限的定量机制：M　条独立信号相加后均值按　M　增长、涨落只按　√M　增长，均值会淹没待提取的信号（另引正性发放率整流的限制）。因此该限制与用何种开关方案无关。

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DISCUSSION

We have proposed an extension of the concept of global balance that offers an alternative to the more traditional model of gating by using inhibitory neurons to deactivate a signaling pathway[14,28,29]. This inverts the usual scheme of allowing signal propagation by default and disrupting the signal flow to gate a pathway off. Instead, in our model, the balanced gated off state is the default. In complex networks, gating presumably occurs in parallel along the many pathways responsible for transmitting different aspects of a stimulus. It may be easier for a system to keep track of what is 'interesting' in a broadband signal stream than to keep track of all the 'uninteresting' stimulus features that should be suppressed. If one feature of a stimulus warrants further processing, a control mechanism can select it by unbalancing its respective module, allowing the signal to propagate further downstream. We used gain modulation to unbalance signal-carrying pathways and gate signals on, but ordinary subtractive inhibition of inhibitory interneurons could also be used in our scheme.

Other discussions of signal switching in balanced neural circuits, either by shifting inhibition or through gain modulation, can be found in refs. 2,19,30. Continuous temporally and strengthwise correlated (balanced) excitatory and inhibitory input activity has been reported *in vivo* through recordings from pairs of pyramidal cells in the rat somatosensory cortex during spontaneous and sensory-evoked activities[31]. It has also been demonstrated that inhibition can be used to adjust the gain of a downstream circuit that receives balanced input and can thus control behavioral responses in a context-dependent manner[32]. These findings fit well into the framework of our hypothesis.

Although we found that gating is robust to several perturbations, it would be interesting to study how a homeostatic mechanism might impose and maintain a detailed balance. Spike timing–dependent plasticity has been shown to generate a global balance between excitation and inhibition[33,34], but detailed balance is likely to require some further competitive as well as homeostatic mechanisms. In developing systems, this might involve tuning AMPA and immature, excitatory GABA synapses to the same degree.

Failure to maintain a precise balance between excitation and inhibition due to various abnormalities of synaptic transmission is commonly hypothesized as a basis for mental disorders such as schizophrenia[35–39] and autism[40,41]. Although it is natural to think that such an imbalance might lead to basic instabilities, such as those associated with epilepsy, it is more difficult to understand how they would lead to cognitive and behavioral disorders. Our results provide a suggestion. If we associate the local inhibitory neurons in our model network with parvalbumin-positive inhibitory neurons in cortex, the failure of gating in this model with reduced inhibition (**Fig. 5c**) could provide a functional basis for the hypothesis that reduced GABA production in parvalbumin-positive interneurons may contribute to gating problems in schizophrenia[35]. Similarly, the inability to discriminate between external and internal activity could be related to the hallucinations and delusions that have been hypothesized to arise from defective dopaminergic regulation[36,37] or NMDA current anomalies[38]. Although these latter anomalies correspond to hypofunction of NMDA conductances, this is associated with a hyperexcitability of the affected circuits[39], so we modeled the overall effect by increasing excitation (**Fig. 5d**).

The mechanism we have proposed makes a distinctive prediction concerning inhibitory activity in a signal-receiving region. Although excitatory neurons receiving a signal should respond more vigorously in an attentive (gating on) than in a nonattentive (gating off) state, at least some local inhibitory interneurons should follow a signal more reliably in the inattentive state and should decorrelate their activity from the stimulus with attention.

In studying the capacity of detailed balance to switch multiple signals, we encountered a fundamental limitation of multisignal encoding using firing rates that affects any switching scheme. However, once this limitation was avoided by keeping too many of the possible signals from being present at any given time, we found a large capacity to gate multiple signals. Thus, detailed balance offers a powerful and dynamic way of controlling signal flow in complex and multiply interconnected circuitry.

METHODS

Neuron model. The model used for all our simulations is a leaky integrate-and-fire neuron, characterized by a time constant, $\tau = 20$ ms, and a resting membrane potential, $V_{\text{rest}} = -60$ mV. Whenever the membrane potential crosses a spiking threshold of -50 mV, an action potential fires and the membrane potential is reset to the resting potential, where it remains clamped for a 5 ms refractory period. To set the scale for currents and conductances in the model, we used a membrane resistance of 100 MΩ.

We modeled synapses onto each neuron as conductances, so the subthreshold membrane voltage obeys

$$\tau \frac{dV}{dt} = (V_{\text{rest}} - V) + g_{\text{ex}}(E_{\text{ex}} - V) + g_{\text{inh}}(E_{\text{inh}} - V) + I_b$$

Reversal potentials were $E_{\text{ex}} = 0$ mV and $E_{\text{inh}} = -80$ mV. The synaptic conductances g_{ex} and g_{inh} are expressed in units of the resting membrane conductance. When the neuron receives a presynaptic action potential, the appropriate postsynaptic variable increases, $g_{\text{ex}} \rightarrow g_{\text{ex}} + \Delta g_{\text{ex}}$ for an excitatory spike and $g_{\text{inh}} \rightarrow g_{\text{inh}} + \Delta g_{\text{inh}}$ for an inhibitory spike. Otherwise, these parameters obey the equations

$$\tau_{\text{ex}} \frac{dg_{\text{ex}}}{dt} = -g_{\text{ex}} \quad \text{and} \quad \tau_{\text{inh}} \frac{dg_{\text{inh}}}{dt} = -g_{\text{inh}}$$

with synaptic time constants $\tau_{\text{ex}} = 5$ ms and $\tau_{\text{inh}} = 10$ ms. I_b is a constant background current used to maintain network activity (see below). The integration time step for the simulations was 0.1 ms. We implemented all simulations in C.

Network architecture. We studied a network of 20,164 leaky integrate-and-fire neurons, laid out on a 142 × 142 grid. Neurons were either excitatory or inhibitory. The ratio of inhibitory neurons was roughly 1 in 4, but the geometric organization of neurons on the grid constrained the final numbers to 15,123 excitatory cells and 5,041 inhibitory cells. Inhibitory neurons were divided into two groups of 3,361 and 1,680 neurons differing in their connectivity pattern. All excitatory neurons and 65% of the inhibitory neurons had a random connectivity of 2% to the rest of the network (red cells in **Fig. 1a**). The 1,680 inhibitory neurons of the second group each targeted 40% of their 500 closest neighbors, thus acting locally[42] (blue cells in **Fig. 1a**). To avoid boundary effects, we implemented the network with the topology of a torus. We used 20,000 cells because this was the largest network that we could study within reasonable computation times. It has been shown previously that the activity in such networks becomes independent of size at about 10,000 neurons[8,11,16]. Other network parameters were chosen in keeping with both general properties of cortical circuits and previous work[11,12,14,16,42].

Signal path. In additional to the general architecture, we introduced a specific pathway from one region of the network to another, which we call sender and receiver subnetworks (**Fig. 1b**). The two subnetworks were chosen to be sufficiently distant from each other to exclude possible interactions through local inhibitory neurons. Synapses from a given excitatory sender neuron were allocated to contact either the excitatory or the locally inhibitory neurons of the receiver region, but not both. This division is made for the sake of tuning simplicity. The numbers of neurons and projecting synapses for the sender subnetwork were chosen to supply each of the ~500 excitatory receiver and ~70 inhibitory receiver neurons with 50 synapses from the sender subnetwork, a number necessary for critical spike propagation with appropriate tuning of

E42 可检验预测：注意时抑制与刺激去相关
作者报告｜与上一条配套：注意状态下局部抑制中间神经元的活动应与刺激去相关。两条合起来给出一个可在同一记录中同时检验的双侧判据，比只看兴奋侧输出更难被其他机制凑巧满足。

E34 作者定位：对传统门控模型的替代方案
作者报告｜作者在讨论中明确把本文定位为传统「用抑制关闭一条默认畅通的通路」模型的替代方案。引用时应把它视为并列的模型方案而非已被否证的旧模型。

E35 方向反转：默认状态是关闭
作者报告｜本文最根本的概念改动是把默认状态反转为「关闭」。作者进一步主张这样也许更容易——追踪信号流中「有趣的」部分比追踪所有「无趣的」待抑制特征更省事。这是一个设计论证，没有成本比较数据支撑。

E48 实现方式的开放性：也可用普通减性抑制
作者报告｜作者说明除了增益调制，用普通的减性抑制关闭抑制性中间神经元同样可以实现打开门控。这扩大了机制的实现空间，也意味着本文的「增益调制」并非必要条件，读者不必把两者绑定。

E46 外部证据支持：平衡在体内有记录
作者报告｜作者引用在体记录，指出大鼠体感皮层锥体细胞对之间存在时间与强度上持续相关的平衡兴奋与抑制输入。这为「平衡是皮层常态而非模型假设」提供了独立支撑，但属于相关性证据，不能替代本文缺失的因果验证。

E44 作者点名的未解问题：稳态机制如何维持详细平衡
作者报告｜作者把「某种稳态机制如何施加并维持详细平衡」列为值得研究的问题。这等于承认本文的详细平衡是被手工设定的属性而非会自己出现的属性，是该机制最主要的开放问题。

E43 已知：STDP 可产生全局平衡
作者报告｜作者指出已有工作表明突触时序依赖可塑性（STDP）能产生 E/I 的全局平衡。这既是本文平衡态可能的自组织来源，也界定了差距：全局平衡已有机制，详细平衡尚无。

Ｅ４５　　作者判断：需要竞争性与稳态机制并行
作者报告｜作者认为详细平衡可能同时需要竞争性机制与稳态机制，即子集级的平衡可能仅靠　ＳＴＤＰ　式的全局平衡不够。这一判断界定了后续机制研究的方向，但本文没有给出任何验证。

Ｅ３６　　病理类比的落点：ＰＶ　中间神经元与门控失败
作者报告｜作者把模型中抑制减弱导致的门控失败与精神分裂症中　ＰＶ　中间神经元　ＧＡＢＡ　产生减少的假说联系起来。这是模型提出的功能性假说，不是本文提供的证据，差异类型应记为证据不足。

Ｅ１１　　网络规模的选取依据
作者报告｜作者说明选两万单元是因为计算时间所限，并用既往「活动在约一万神经元时已与规模无关」的结果为「再大也不会有新现象」提供依据。这属于对模拟外推范围的合理性论证，不是本文自身的敏感性分析。

Ｅ０９　　关键结构假设：源神经元只投　Ｅ　或只投　Ｉ
作者报告｜方法部分暴露的核心简化：每个兴奋性源神经元的突触被分配为只接触接收区兴奋神经元或只接触局部抑制神经元，绝不两者兼顾。这使得「按通路打开抑制支路」在工程上可行，但它把真实皮层中同一投射同时到达　Ｅ　与　Ｉ　的常见情形排除在外。

Ｅ１０　　作者自述：该假设出于调参简便
作者报告｜作者明说这一结构性限制是为调参简便而作，不是从生物学依据推出的。这一点决定了本文机制是「在特定构型中可行」的证明，而不是「皮层就是这样连」的论证。

Ｅ４１　　可检验预测：非注意时抑制神经元更跟随刺激

作者报告｜这是全文最可借用的一条在体可测签名：门控关闭（非注意）状态下，接收区的局部抑制中间神经元应当比门控打开状态更可靠地跟随刺激。它与「注意提升抑制神经元响应」的直觉方向相反，因此不易被无意满足。

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synaptic strengths (see below). In addition, the number of neurons in both subnetworks was chosen to be as large as possible without interfering with overall network activity during signal propagation[14]. The final numbers were 494 excitatory–excitatory sender neurons, 234 excitatory–inhibitory sender neurons, 463 excitatory receiver neurons and 73 inhibitory receiver neurons.

Tuning conditions. Except for synapses along the signaling pathway and those mentioned further below, all synapses of the same cell type had the same strength. A background current (I_b) of 0.03 nA was delivered to every neuron and the three sets of strengths were adjusted to allow asynchronous background activity within the network. The postsynaptic conductances of $\Delta g_{ex}^{global} = 0.8$ nS, $\Delta g_{inh}^{local} = 1.5$ nS and $\Delta g_{inh}^{global} = 7.5$ nS correspond to 0.5 mV excitatory postsynaptic potentials and −0.4 mV and −1.1 mV inhibitory postsynaptic potentials, respectively, as obtained from spike-triggered averages in the active network. To propagate signals from the sender subnetwork to the excitatory receiver subnetwork, we set the synapses between those two groups to $\Delta g_{ex\to ex}^{sender\to receiver} = 0.9$ nS. The synapses between sender neurons and inhibitory receiver neurons were left unchanged, and the inhibitory synapses between the inhibitory and the excitatory neurons in the receiver subnetwork were strengthened to $\Delta g_{loc\ inh\to ex}^{receiver\to receiver} = 4.65$ nS. The extra tonic excitation that the inhibitory receiver neurons receive from background activity through the projections from the sender subnetwork was compensated by strengthening global inhibition to these cells to $\Delta g_{inh\to loc\ inh}^{net\to receiver} = 9.4$ nS. Under these conditions the system is sufficiently balanced to prevent correlated inputs in the sender subnetwork from modulating the firing pattern of the excitatory receiver subnetwork.

To allow propagation, the balance between the excitatory and the inhibitory signal was modified by decreasing the gain of the inhibitory receiver neurons. In integrate-and-fire neurons such a gain change is equivalent to reducing the strength of all synapses by 85% in the case of symmetric gain changes and by 30% in the asymmetric case in which only the response amplitude to excitatory inputs is altered. These values were chosen to minimize similarity values in the gated-off state (**Fig. 4a**).

To compensate for cell loss (**Fig. 4c**), we calculated the decrease of overall synaptic strength and redistributed the difference equally among the remaining synapses in the pathway. The gating mechanism can function with only 20 inhibitory receiver neurons when we strengthen their synapses threefold, to 15 nS (approximately the same strength as two globally inhibitory neurons), and allow them to fire at rates of up to 600 Hz (**Fig. 4d**).

Two input signals. To propagate and control an additional signal to the receiver region, a second inhibitory receiver subnetwork is necessary. Some of the globally acting inhibitory neurons in the receiver region can be recruited as locally inhibitory for that purpose by generating a new local architecture for them. An additional set of Poisson input spike trains is generated and connected synaptically to both the shared excitatory receiver subnetwork and the new inhibitory receiver subnetwork. As before, the synapses of the Poisson population are tuned to drive their respective target subnetworks in the absence of additional correlated signal input. To balance two active signals at the same time, some of the synapses within the network must also be retuned. See **Table 1** for a complete listing of all modified synapses.

Pathologies. We chose two ways to disrupt the detailed balance mechanism (**Table 1**). First, we introduced a deficiency in the locally inhibitory neurons, including those of the inhibitory receiver subnetwork, by decreasing their synaptic strengths to $\Delta g_{inh}^{local} = 0.6$ nS. Second, we induced a hyperexcitability of the receiver region by increasing all the excitatory synapses from the rest of the network onto the receiver neurons to $\Delta g_{ex\to ex}^{net\to receiver} = 1.28$ nS and $\Delta g_{ex\to loc\ inh}^{net\to receiver} = 1.28$ nS. These two manipulations are independent and can be combined without retuning.

Multiple input signals to a single cell. In the later part of the paper, we modeled the gating of multiple signals in a single integrate-and-fire cell that receives 800 excitatory and 200 inhibitory synapses modeled as Poisson processes with temporally changing spiking probabilities. To avoid unrealistically high membrane currents when many inputs arrive at the cell, synaptic strengths were tuned down to $g_{ex} = 0.014$ nS and $\Delta g_{inh} = 0.044$ nS with postsynaptic potentials of 0.13 mV and −0.22 mV at V_{rest}, respectively. When we drove each afferent with more than one signal, the overlap (effectively a summation of spiking probabilities for each synapse) demanded a rescaling of the input rates to a dynamic signal range between 0 and 150 Hz.

The following procedure is used to compute the inhibitory gain factors needed to gate 'on' one signal among many. First, to specify which signal is connected to which inhibitory afferent, we define an $N \times M$ matrix B with $B_{ia} = 1$ if inhibitory afferent i receives signal a and $B_{ia} = 0$ if it does not. The total inhibitory input due to signal a is then $C_a = \sum_i B_{ia}$. We next choose a particular signal, say signal 1, to gate on. We do this by setting the gains for all the inhibitory neurons receiving signal 1—that is, all neurons with $B_{i1} = 1$—to zero. We then adjust the firing rates (gains) of the remaining inhibitory afferents to compensate for these missing afferents for all other signals (missing because their gains are zero). If n afferents receive signal 1, we define an $(N-n) \times (M-1)$ matrix $\tilde{B}$, which is just B with signal 1 and all of the afferents connected to signal 1 removed. Define $\tilde{C}$ to be a vector with $M-1$ components given by $\tilde{C}_a = C_{a+1}$ for $a = 1, 2, \ldots, M-1$. The inhibition missing because the afferents receiving signal 1 have been turned off can be replaced if the firing rates of the afferents not receiving signal 1 are multiplied by a row vector of gain factors α such that $\alpha\tilde{B} = \tilde{C}$. This equation is 'solved', in the sense of minimizing the square of the difference between the two sides summed over a, by setting $\alpha = \mathrm{pinv}(\tilde{B})\tilde{C}$, where $\mathrm{pinv}(\tilde{B})$ is the pseudoinverse of $\tilde{B}$. The gains of all inhibitory afferents receiving signal 1, by contrast, are set to zero. This determines the complete set of gains used to gate signal 1 on and leave all other signals off. To gate signal 1 off, all the gains are set to 1. A similar procedure is used for any other signal that we wish to gate.

Response properties of the balance mechanism. To supply a signal to the network, we generated Poisson input spike trains with a firing rate $r_0(t)$ as a source of input to the network. Each input spike generated by that group increased the excitatory synaptic conductances in neurons of the sender region by $g_{ex} \to g_{ex} + \Delta g_0$. The synaptic strength Δg_0 was tuned so that the firing rates of the sender neurons reproduce the input signal, that is, they track the input firing rate $r_0(t)$.

We used a correlation measure[12,14] to determine how similar the firing rates in the receiver region were to the input. To do this, we calculated the population firing rate $r(t)$ in 5-ms bins by counting spikes and also determined its time-averaged value $\bar{r}$. The correlation is then

$$C(\tau) = \frac{\langle (r_0(t) - \bar{r}_0)(r(t+\tau) - \bar{r})\sigma_{r_0} \rangle_t}{\langle (r_0(t) - \bar{r}_0)(r_0(t+\tau) - \bar{r}_0)\sigma_r \rangle_t}$$

where the brackets denote an average over time, $r_0(t)$ and $\bar{r}_0$ are the firing rate and average for the input, and σ_r and σ_{r_0} are the s.d. values of the corresponding firing rates. We used the activity of the input as a reference rather than the sender subnetwork to distinguish signal transmission from propagation of fluctuations arising in the sender. Signal propagation between subnetworks is then characterized by reporting the maximum value (over τ) of $C(\tau)$, which we call the similarity[14]. For the analysis of multisignal gating in a single integrate-and-fire cell, we used a similarity measure that was not normalized by σ_r to avoid overestimating the quality of the output signal in cases when the output firing rate was greatly diminished.

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E15 调参目标：门控关闭时信号必须传不过去
AI解读｜作者说明整套突触强度是围绕这一目标调的。这说明「关得住」不是自然涌现的性质，而是被主动优化出来的前提；因此后文关于「打开」的结论都继承了这一前提的强度。

E12 打开门控的操作定义
作者报告｜方法部分给出打开门控的可操作定义：降低抑制性接收神经元群的增益。注意作者并未实现任何具体的细胞内增益机制，因此「增益调制」在本文中是一个通过参数缩放施加的外部操作。

E13 增益调制只是突触强度的代理
作者报告｜作者说明在 LIF 模型中增益下调等价于把突触强度整体降低 85%（非对称情形降 30%）。这揭示了一个重要实现边界：本文并未模拟胆碱能或膜电导调制这类真实的增益控制，机制的可迁移性受此限制。

E14 相似度指标以输入为参考
作者报告｜作者有意以输入而非信号源区作为互相关参考，理由是要把「信号被真正传输」与「信号源自身的涨落被传下来」区分开。这使相似度成为衡量内容传输而非活动搬运的合适指标，也是本文评价体系设计得较好的一处。

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